

# Saumya Bhandari

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## TECHNICAL SKILLS

**Core Languages:** Python (Advanced), C++, C, Swift (iOS Integration), SQL, Bash

**Generative AI Multi-Agent Systems:** Google ADK, Model Context Protocol (MCP), Vertex AI (Gemini 2.5/3.5), Multi-Agent RAG, GraphRAG, Qdrant, Pinecone, LoRA/QLoRA Fine-tuning, LangChain

**Backend Distributed Systems:** Google Cloud Tasks, Distributed Lease Locking, Rate Limiting, FastAPI (Async), Redis, Celery, MongoDB (ACID Transactions), PostgreSQL (asyncpg), WebSockets, Kafka

**Computer Vision (Edge 3D):** PyTorch, TensorRT, CUDA, DeepStream, YOLOv8/World, 3D PnP, 2D-3D Homography, OpenCV, Stable Diffusion, ControlNet, SegFormer

**MLOps Cloud Infrastructure:** Kubernetes, Docker, CI/CD, AWS (SageMaker, Lambda, S3, Textract), GCP, Model Quantization (INT8), Jetson Orin

## CORE ENGINEERING EXPERIENCE

### • Senior AI Engineer

*Giles.app (United Kingdom)*

Mar 2026 – Present

*Remote*

- Architected an asynchronous **LLM Queue middleware** decoupling Google ADK agents from Vertex AI (Gemini 2.5/3.5) via Google Cloud Tasks (4 dispatches/sec, DLQ sweeper); engineered stateful **QueueItem** and **QueueItemStep** audit ledgers across MongoDB replica sets with ACID transactions and 30-min selective TTL scrubbing.
- Designed a hierarchical multi-agent tree (**GilesChatAgent** → **ResearchOrchestrator** → Domain/Leaf Agents); re-engineered system prompts with dynamic turn-scoped contexts, semantic constraint injection, and adjective disambiguation (conversational vs. UK NICE) for zero-deadlock, deterministic routing.
- Engineered declarative **MCP tool servers** across European HTA bodies (**IQWiG, G-BA, HAS, NICE**); implemented **scoped agent** execution and pioneered **connector coupling** (*GermanHTAGroup*) to eliminate candidate drug divergence in AMNOG workflows while preserving parallel search.
- Built a fault-tolerant, resumable document ETL pipeline in **Qdrant** (Blue-Green point swaps, quarantine isolation, ETag/304 cache limits) with a **cluster-global permit scheduler** ( $\geq 3000\text{ms}$ ,  $\leq 1,000\text{ req/hr}$ ); accelerated queries by **parallelizing multi-source comparative RAG analyses**.

### • AI Engineer III (Promoted)

*Accuwiz Tech – Industry AI*

Jul 2025 – Feb 2026

*Kathmandu / Remote*

- **Signal Healthcare Ecosystem (SignalDX.ai):** Architected a multi-service clinical lab intelligence ecosystem delivering 4 production platforms: **SignalDX & SignalDX-MCP** (engineered a **Neo4j-based GraphRAG** policy graph indexing **10,000+ CMS LCD/NCD coverage rules** across 14 node types with multi-hop Cypher traversals linking ICD-10 indications to ordered CPT/HCPCS lab panels to eliminate billing denials; integrated an autonomous *MedicareAgent* over live Model Context Protocol [MCP] streamable sessions and Stedi eligibility APIs); **SignalDoc** (stateless microservice classifying **22 canonical clinical document types** via Vision LLMs with 100% in-memory zero-disk multipart buffering [HIPAA-compliant zero /tmp rollover], cryptographic SHA-256 provenance, and strict Pydantic extraction); and **SignalMD** (functional medicine engine extracting **100+ biomarkers** across 13 panel types with LOINC standardization and Redis session caching).
- **Spatial Computer Vision & Blueprint Systems (Cabinet Design & Lightology):** Engineered and deployed 2 physical-world spatial intelligence engines before client handoff: **Lightology** (physically scaled lighting visualization decoupling scene geometry from generative fusion—segmented dining/island anchors via Transformer/SAM2, calculated vanishing-point perspective homography for Pixels-Per-Inch [PPI] calibration, and fused millimeter-accurate fixture cutouts with iterative GrabCut fallbacks); and **Cabinet Design** (monorepo CV engine unwarping perspective kitchen photos into 2D orthogonal elevation blueprints—fused instance masks, depth maps, and 3D surface normal vectors to cluster planar walls [*PlaneDeterminer*], then applied homography rectification [*PlaneHomographyRectifier*] and standard US modular sizing [9"–48" widths, 34.5" base heights] with Celery/Redis workers and robust bounding-box fallback clustering).

### • AI Engineer II

*Accuwiz Tech – Industry AI*

Nov 2024 – Jun 2025

*Kathmandu / Remote*

- **Virtual Staging Platform (PadStyler.com):** Delivered the core generative vision engine across **11 multi-turn renovation modalities** (virtual staging, item removal, floor/wall refurbishing, kitchen/bath/backyard remodels, video generation); engineered depth-aware (MiDaS) scene parsing combining **fine-tuned Transformer-based segmentation pipelines (SegFormer/ADE20K)** and **CNN-based architectures (YOLOv8/v11)**; extracted high-fidelity **segmented binary masks** to perform boolean mask algebra (union, difference, XOR, intersection) and morphological transformations, strictly locking structural room boundaries (moldings, windowpanes, baseboards). Productionized on AWS FastAPI with Redis session management, migrating streaming delivery to secure **S3 Presigned URLs** with ThreadPool semaphore concurrency control.
- **Ordaana Voice AI (Ordaana.com):** Engineered and deployed real-time voice-to-voice AI receptionist infrastructure powered by xAI Grok Voice API, Twilio, and FastAPI; built full-duplex WebSocket audio streaming achieving **400–800ms** first-response latency with

dynamic runtime company context injection, pre-speech command queuing, multi-session state isolation, automated CRM calendar booking (Google/Cal.com), structured lead capture, and ChatGPT-style resumable session persistence with SQLite/SQLModel. Completed live client pilot leading to Level III promotion.

- **Career Break** May 2024 – Oct 2024  
*Personal Medical Recovery* Kathmandu, Nepal
  - Took a planned medical leave for personal health recovery and rehabilitation before returning to full-time engineering.
- **Machine Learning Engineer** Jan 2024 – Apr 2024  
*Naamche (ReAlpha) – PropTech AI* Kathmandu, Nepal
  - Virtual Staging Platform: Fine-tuned SegFormer for room segmentation; diffusion-based inpainting pipeline increased buyer-engagement metric **86%**; served on SageMaker serverless multi-GPU endpoint.
  - Collaborated to deliver “Claire” – RAG-based virtual real-estate agent and advisor (GPT-4 + Pinecone) answering complex buyer QAs.
- **Machine Learning Engineer** Sep 2022 – Oct 2023  
*Wiseyak Inc. – Medical AI*
  - Delivered chest X-ray report generator **AUROC: >0.91** (multi-label classification) on highly imbalanced data; Grad-CAM explainability integrated into radiologist cockpit. Extended system on GPT-2 report generation platform using CLIP embedding fine-tuning.
  - Designed CNN-Transformer hybrid for medical video denoising + next-frame prediction; custom Convolutional Transformer with Transformer encoder for temporal understanding.
  - Built domain-adversarial network (DANN) using EfficientNet for plant disease detection aligned with Meta’s SAM for reducing domain shift.

## TECHNICAL ADVISORY & CONSULTING ENGAGEMENTS

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- **Consulting AI Engineer** Oct 2024 – Feb 2026  
*InstaComply – Legal-Tech / Compliance* Contract
  - Architected **SelfCompl.ai** – a modular multi-agent RAG system powering GDPR, ISO-27001, and SOC2 audits; engineered compliance-as-code infrastructure as a real-time compliance gateway for Fintech enterprises.
  - Designed the core AI blueprint: vector DB routing, human-in-the-loop (HITL) feedback loops, XAI audit trails, and cost-optimized multi-LLM orchestration (Claude, GPT-4o, Gemini, Mixtral).
  - Authored 5+ research-backed articles on AI regulatory compliance and automated governance published at [selfcomplai.com/blogs.html](https://selfcomplai.com/blogs.html).
  - Stack: Python, FastAPI, PyTorch, LangChain, Pinecone, AWS Textract, PostgreSQL RDS.
- **AI and Computer Vision Advisor** Jul 2025 – Jul 2026  
*PointRef Inc. (DE C-Corp) – AI Sports-Tech* Advisory

Advising: [Mosalam Ebrahimi](#) (ex. Meta Reality Labs) | [pointref.com](https://pointref.com)

  - Engineered the core computer-vision architecture for real-time pickleball officiating: formulated monocular 2D-to-3D **Perspective-n-Point (solvePnP/Ransac)** camera calibration and pose estimation mapping frames to a metric 20 × 44 ft court; derived planar **homography estimation** (RANSAC/MAGSAC) for sub-inch court registration and reprojection error minimization.
  - Conducted heavy dataset curation and augmentation (Mosaic, Mixup, Albumentations) to train custom YOLO and ResNet ball and keypoint detectors; engineered physics-grounded **bounce detection algorithms** (parabolic trajectory fitting, vertical velocity sign flips, local extrema, and paddle-contact rejection) coupled with dynamic multi-frame **ball trails** for millimeter-accurate line calling.
  - Converted PyTorch models to **Core ML** for on-device edge inference; integrated real-time tracking, rally metrics, speed estimation, and serve analytics into the native Swift iOS app ([app\\_b2c](#)) featuring offline-first direct-to-cloud upload pipelines, successfully published on the **Apple App Store / TestFlight**.

## EDUCATION

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- **B.Sc. (Hons) Computer Science (First-Class)** 2019 – 2022  
*Herald College Kathmandu, Naxal, Kathmandu*
  - Thesis: *Pathfinder: An Image Segmentation-Based Navigation System for Self-Driving Systems*
  - Student Academic Representative (STAR) **2019–2021** (2 Years) | **Hult Prize 2021** On-Campus Winner
  - Actively founded and led 3 student-run technical communities

## SELECTED CONTINUOUS SELF-LEARNING INTERESTS

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- **Stanford:** CS231N (CNN), CS224N (NLP) | **MIT:** 6.034 (AI), 18.06 (Linear Algebra), Deep Learning Bootcamp, 9.13 (Human Brain), 8.04 (Quantum I) | **Harvard:** Stat110 | **UW:** CS480/680 (Deep Learning)